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High-performance annealing-free wurtzite ferroelectric field-effect transistors with ultrathin oxide semiconductors for neuromorphic computing

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Wurtzite-structured aluminum scandium nitride (AlScN) ferroelectric materials are potential candidates for non-volatile memory (NVM) applications because of their high remanent polarization (P_r > 100 μC·cm^{-2}) and compatibility with back-end-of-line (BEOL) processes. However, achieving high-performance wurtzite ferroelectric field-effect transistors (FeFETs) remains a significant challenge because of their high interface trap density (D_{it}) and large threshold voltage (V_{TH}) variation. Here we show the distinct polarization switching mechanisms of AlScN FeFETs based on an ultrathin indium gallium oxide (IGO) channel with three different structures. FeFETs with a metal–ferroelectric–metal–insulator–semiconductor (MFMIS) structure show high interface quality with low D_{it} and a stable V_{TH}, achieving a large memory window (MW) of up to 20.6 V and a high endurance of 10^6. The stable 6-bit operation of our devices with 64 distinct states shows the feasibility of multilevel storage applications. Furthermore, we demonstrate the applicability of the FeFET in neuromorphic networks, with a high recognition accuracy of 93.02%.

The exponential growth of data-centric applications, such as artificial intelligence, edge computing, and the Internet of Things, has exposed the inherent limitations of traditional von Neumann architecture, where the physical separation of processing and memory units leads to severe 'memory wall' and 'power wall' bottlenecks[1–5]. Ferroelectric field-effect transistors (FeFETs) have emerged as promising candidates for overcoming these bottlenecks because of their inherent non-volatile memory (NVM) capability and in-memory computing (IMC) potential. By integrating ferroelectric materials into the gate stack, FeFETs realize the fusion of memory and logic functions at the single-device level through polarization-induced threshold voltage modulation, which not only enables high-speed, low-power NVM operations but also mimics synaptic weight plasticity for neuromorphic computing applications[6–10].

In recent years, FeFETs have undergone significant advancements in material systems, device architectures and applications[11]. However, FeFETs still face critical challenges that hinder their practical implementation and commercial deployment. For the most extensively studied hafnium oxide (HfO_2)-based FeFETs, the remanent polarization (P_r) is relatively low (10–40 μC cm^{-2}), leading to an insufficient

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memory window (MW)[12]. Moreover, reliability issues such as wake-up and fatigue effects still exist, which limit retention and endurance and induce variability[13]. Additionally, most HfO_2-based FeFETs require high-temperature post-annealing treatment to stabilize the ferroelectric phase, which increases the process complexity and energy consumption and is incompatible with back-end-of-line (BEOL) integration[14,15]. The discovery of ferroelectricity in wurtzite aluminum scandium nitride (AlScN) in 2019 opened an avenue for high-performance FeFETs[16]. Compared with HfO_2-based ferroelectric materials, AlScN has distinctive advantages (Fig. 1a)[17–22]. First, AlScN possesses a square-shaped polarization-electric field (P–E) loop with a large tunable coercive field (E_c) of 3–6 MV cm^{-1} and an ultrahigh P_r exceeding 100 μC cm^{-2}, which is more than three times greater than that of HfO_2-based materials[23–25]. Second, AlScN maintains stable ferroelectricity with a high Curie temperature (T_C) exceeding 1100 °C, far outperforming HfO_2-based materials in terms of thermal stability[26–28]. Most importantly, AlScN is highly CMOS BEOL compatible and annealing-free at low process temperatures (<400 °C) and has a single, stable wurtzite ferroelectric phase, unlike HfO_x polymorphisms, which exhibit ferroelectricity only in metastable phases[29–31]. On the other hand, amorphous oxide semiconductors (AOSs) grown by atomic layer deposition (ALD) are considered competitive channel material candidates because of their wide bandgap, high carrier mobility, and low leakage current, presenting high-performance BEOL-compatible electronic devices for monolithic 3D integration[32,33]. Compared with other deposition techniques, such as magnetron sputtering, ALD offers self-limited growth, atomic-level film uniformity and ultrathin channel capability (down to a few nanometers)[34,35]. Among the AOSs used for channel materials of FeFETs, ultra-thin indium gallium oxide (IGO) grown by ALD[36] has high carrier mobility and better stability than other AOSs[37,38]. Although some AlScN FeFETs with a metal-ferroelectric-semiconductor (MFS) structure have been demonstrated, to the best of our knowledge, no available approach has been reported to effectively improve the interface quality and V_{TH} variation and thereby achieve high-performance AlScN-based FeFETs with ALD ultra-thin oxide semiconductors.

In this work, we report a high-performance and annealing-free FeFET memory device based on AlScN and ALD ultra-thin IGO channels with an MFMIS structure. The introduction of a high-quality HfO_2 interlayer (IL) shifts the interfacial mechanism from a charge-trapping-dominated regime to one governed primarily by ferroelectric switching and the floating gate greatly decreases the threshold voltage variation. AlScN-based FeFETs with an MFMIS structure have a large MW of up to 20.6 V, a high ON/OFF ratio of over 10^7, excellent endurance over 10^6 cycles, and extrapolated retention beyond 10 years. Additionally, AlScN MFMIS FeFETs exhibit favorable performance in terms of both multi-level memory (MLM) and neuromorphic synapse emulation, thus demonstrating great promise for high-performance NVM and IMC technologies.

Results and discussion

Ferroelectric properties of the in situ AlScN capacitors

To investigate the effect of the sputtering atmosphere on the ferroelectricity of Pt/AlScN/Pt metal–ferroelectric–metal (MFM) capacitors, we fabricated two kinds of AlScN films via reactive magnetron sputtering under Ar/N_2 (N_2 samples) and Ar/N_2/H_2 (N_2/H_2 samples) atmospheres. Figure 1b shows the polarization-electric field (P–E) and current density-electric field (J–E) hysteresis loops of the N_2 samples measured under 10 kHz bipolar triangle waves. The curve clearly exhibits positive-negative asymmetry, with positive polarization (P_{sw+}) attaining a high value of 166 μC cm^{-2} and negative polarization (P_{sw-}) of 131 μC cm^{-2}. In addition, the peak value of the positive current density is nearly twice the magnitude of the negative current density. Figure 1c shows the P–E and J–E curves at 10 kHz of the N_2/H_2 samples. In contrast to those of the N_2 samples, the P_{sw+} and P_{sw-} of the N_2/H_2 samples achieve similar values of ~147 μC cm^{-2}. Since ferroelectric switching dynamics, including the coercive electric field (E_c) or leakage current, depend on frequency, to further compare the leakage current of both

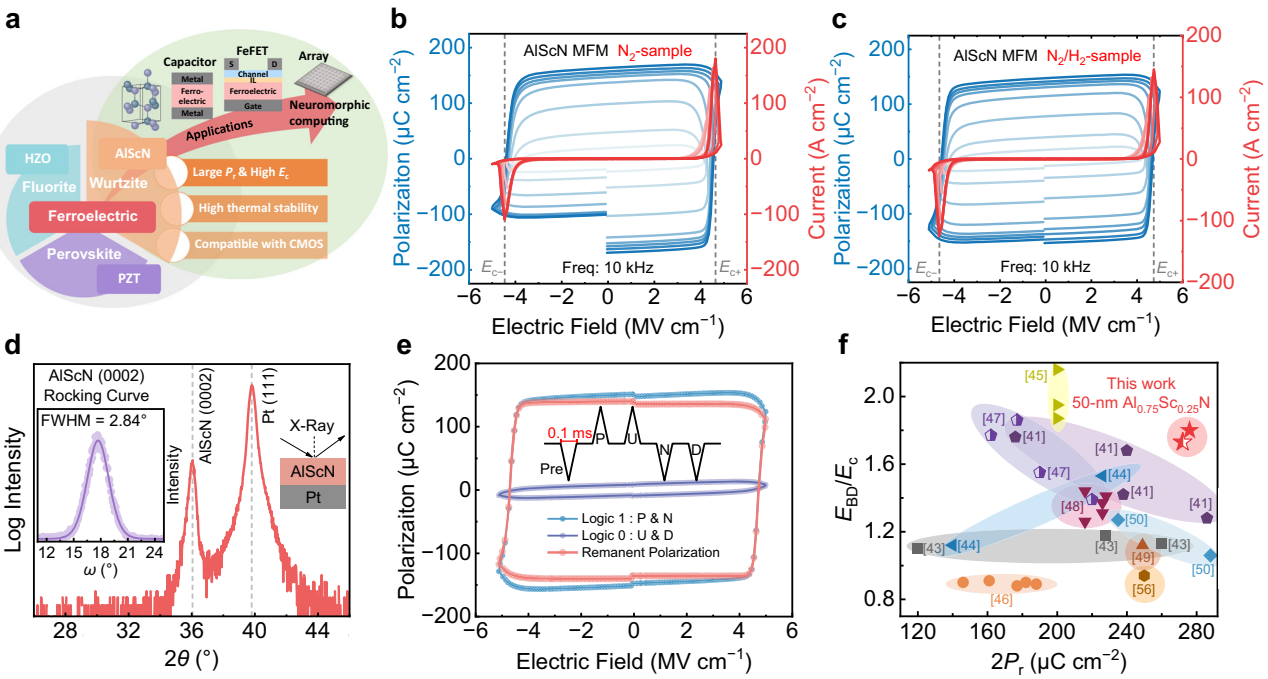


Fig. 1 | Ferroelectric properties of the in situ AlScN capacitors. a The development and applications of ferroelectric materials and the advantages of wurtzite AlScN. P–E and J–E curves at 10 kHz of AlScN MFM capacitors sputtered with **b** N_2 and **c** N_2/H_2, respectively. The dashed lines indicate the E_c extracted from the J–E curves. **d** Out-of-plane $\theta-2\theta$ XRD pattern of AlScN films on Pt template. The inset shows the rocking curve corresponding to the AlScN (0002) peak. **e** PUND measurements of the N_2/H_2 sample at 10 kHz. **f** Benchmark of E_{BD}/E_c versus $2P_r$ of reported AlScN-based MFM capacitors.

samples, we measured J–E curves at 1 kHz (Supplementary Fig. 3). Similarly, the values of the leakage current extracted at E_c of the N_2/H_2 samples are approximately −0.53 and +0.28 A cm^{-2}, which are much lower than those of the N_2 samples. These results indicate that the introduction of H_2 during sputtering effectively mitigates the polarization asymmetry of the fabricated AlScN films. According to previous studies on the switching dynamics of AlScN[39,40], the polarization asymmetry may be associated with the imprint effect or the spatially non-uniform distribution of electrically active defects. In addition, first-principles calculations have established that native defects such as nitrogen vacancies and oxygen-related defects introduce gap states in AlScN and can strongly affect both leakage transport and polarization-switching energetics[41]. Therefore, we believe that the effect of H_2 should be understood as a passivation of oxygen-related defects and their associated charge distribution, which can reduce the spatial non-uniformity of the internal electric field, thereby weakening the imprint responsible for the different switching fields under positive and negative polarization switching. In parallel, the reduced population of electrically active defect states can suppress trap-assisted carrier transport, leading to lower leakage of N_2/H_2 samples. Consequently, the AlScN films used in the FeFETs in this work were all sputtered under the same conditions as the N_2/H_2 samples because the N_2/H_2 samples have more symmetrical polarization and lower leakage.

Figure 1d presents the X-ray diffraction (XRD) pattern of the N_2/H_2 sample. The peaks at 36° and 39.8° correspond to AlScN(0002) and Pt(111), respectively, indicating the good crystallinity of the AlScN film grown on the Pt template. The inset shows the rocking curve corresponding to the AlScN(0002) peak; the full width at half maximum (FWHM) of 2.84° indicates a well-textured c-axis orientation of the fabricated AlScN films. The root-mean-square (RMS) roughness of the AlScN films measured via atomic force microscopy (AFM) is only 0.46 nm, indicating a high-quality surface for device application (Supplementary Fig. 1). To evaluate the ferroelectric polarization and leakage behavior in the AlScN film more accurately, the positive-up-negative-down (PUND) method with 10 kHz triangle pulses is employed for remanent polarization (P_r) measurements. The logic 1 loop represents a hysteresis measurement in which both pulses (P and N) are switching the sample. In contrast, the logic 0 loop contains only non-switching polarizations (U and D). Subtracting logic 0 from logic 1 provides the remanent-only loop. The PUND measurement of the N_2 sample reveals that an asymmetric polarization is obtained with a positive remanent polarization (P_{r+}) of 151 μC cm^{-2} and a negative remanent polarization (P_{r-}) of 121 μC cm^{-2} (Supplementary Fig. 4), whereas the P_{r+} and P_{r-} of the N_2/H_2 sample obtained via this method is similar at 136 and 140 μC cm^{-2}, respectively (Fig. 1e). The symmetry of the ferroelectric polarizations of both samples observed in the PUND test is consistent with the P–E measurement. Supplementary Fig. 5 presents a comparison of P_{sw} values extracted from P–E measurements at various electric fields. Supplementary Fig. 6 shows the direct current (DC) characterization and breakdown electric field (E_{BD}) of both samples. These results reveal a lower E_c asymmetry and a higher E_{BD} for the N_2/H_2 samples than for the N_2 samples[42,43], supporting the conclusion that H_2 suppresses the electrically active and spatially asymmetric defects of AlScN films. For normalization, Fig. 1f presents the benchmark of the E_{BD}/E_c ratio versus $2P_r$ (= $P_{r+} + P_{r-}$) of reported AlScN-based MFM capacitors[42,44–52], showing the great polarization and reliability of our well-optimized AlScN films with H_2 incorporation. Detailed information on this benchmark is provided in Supplementary Table 1.

Effect of the interlayer on the device behavior

On the basis of the high P_r and low leakage AlScN films, we subsequently fabricated different gate stacks of AlScN/IGO FeFETs with (metal–ferroelectric–insulator–semiconductor (MFIS)) HfO_2 layers and without (metal–ferroelectric–semiconductor (MFS)) HfO_2 layers to investigate the effects of the ILs on the device characteristics. Figure 2a, b present a schematic view of an MFIS FeFET and a transmission electron microscopy (TEM) image of the MFIS gate stack, which reveals the high crystallinity inside the AlScN film. Notably, a 3-nm-thick native oxide layer formed on the surface of the 50-nm-thick AlScN due to the ex-situ process. Supplementary Fig. 7 presents the cross-sectional elemental analysis from energy dispersive X-ray spectroscopy (EDS), which reveals that the atomic fraction ratio of Al to Sc is nearly 3:1, which indicates that the Sc concentration (x) in $Al_{1-x}Sc_xN$ is 0.25. Supplementary Fig. 8 presents X-ray photoelectron spectroscopy (XPS) and AFM images of ALD-grown IGO. An ultralow RMS roughness value of 0.12 nm reveals great film uniformity of the IGO channel. Figure 2c presents a picture of the FeFET array and a scanning electron microscope (SEM) image of a 100-nm channel length (L_{ch}) device. Figure 2d shows the capacitance–voltage (C–V) curves of the capacitors of the MFM and MFIM gate stacks corresponding to the device structure. MFIM capacitors with the HfO_2 IL have a lower capacitance and a higher E_c. The clear butterfly shaped curves indicate that the ex-situ capacitors still retain clear ferroelectricity. Figure 2e shows the experimental dual-sweep transfer characteristic curves of the MFS and MFIS devices at V_G = ±15 V and V_D = 0.05 V, which exhibit clockwise and counterclockwise hysteresis, respectively. Furthermore, these results are independent of the channel length of the devices (Supplementary Figs 9 and 10), indicating that the intrinsic differences in charge dynamics between the two gate stacks account for this phenomenon[53]. Supplementary Fig. 11 shows the transfer characteristics of both types of devices in the low gate voltage (V_G) range and a comparison of the extracted subthreshold swing. Under a low gate voltage, where the ferroelectric is barely switching, the MFS device retains a distinct degree of clockwise hysteresis, whereas the counterclockwise hysteresis of the MFIS device is almost negligible. Moreover, the MFIS device features a high ON/OFF ratio of 10^8 and a low subthreshold swing approaching the theoretical limit, indicating a favorable channel/dielectric (IGO/HfO_2) interface with low trap density. Supplementary Fig. 12 presents the low-frequency (1/f) noise measurements of the MFS and MFIS FeFETs. The interface trap density (D_{it}) extracted from the MFIS device is one order of magnitude lower than that of the MFS device. Figure 2f shows the simulated transfer curves of the two types of devices with substantially different D_{it} values, which exhibit completely opposite hysteresis directions, which are consistent with the experimental results. Detailed information on the simulation is provided in Supplementary Note 1. These results indicate that the opposite trend of hysteresis is strongly related to the different interfacial qualities. Figure 2g, h show the energy band diagrams of the MFS and MFIS devices operating at different gate voltages. Owing to the presence of a native oxide layer with many interfacial defects, channel carriers can easily be trapped. The carrier-deficient channel leads to a significant increase in the screening length (λ) and depolarization field (E_{dep}) of the AlScN layer, suppressing the manifestation of ferroelectric polarization[54,55]. Therefore, charge trapping is dominant when the channel directly contacts the surface-oxidized AlScN. During the dual sweep of V_G, channel carriers repeatedly experience trapping at positive bias and detrapping at negative bias, thus leading to the formation of clockwise hysteresis[53]. By introducing a high-quality HfO_2 IL, not only can the trapping of carriers by interfacial defects be suppressed but also sufficient charges ($Q_{FE/IL}$) can be provided to decrease E_{dep}, which enables ferroelectric polarization to be dominant and results in counterclockwise I_D–V_G curves. Moreover, the thin HfO_2 interlayer can protect AlScN from further oxidation during subsequent ALD IGO channels.

Effect of the floating metal on the device behavior

To enable high storage density and accurate multilevel cell (MLC) operation, the large MW and low V_{TH} variation of the FeFET are considered highly desirable. Figure 3a shows a schematic of MFMIS-stack FeFETs with Pt floating metal layer insertion. Supplementary Fig. 14

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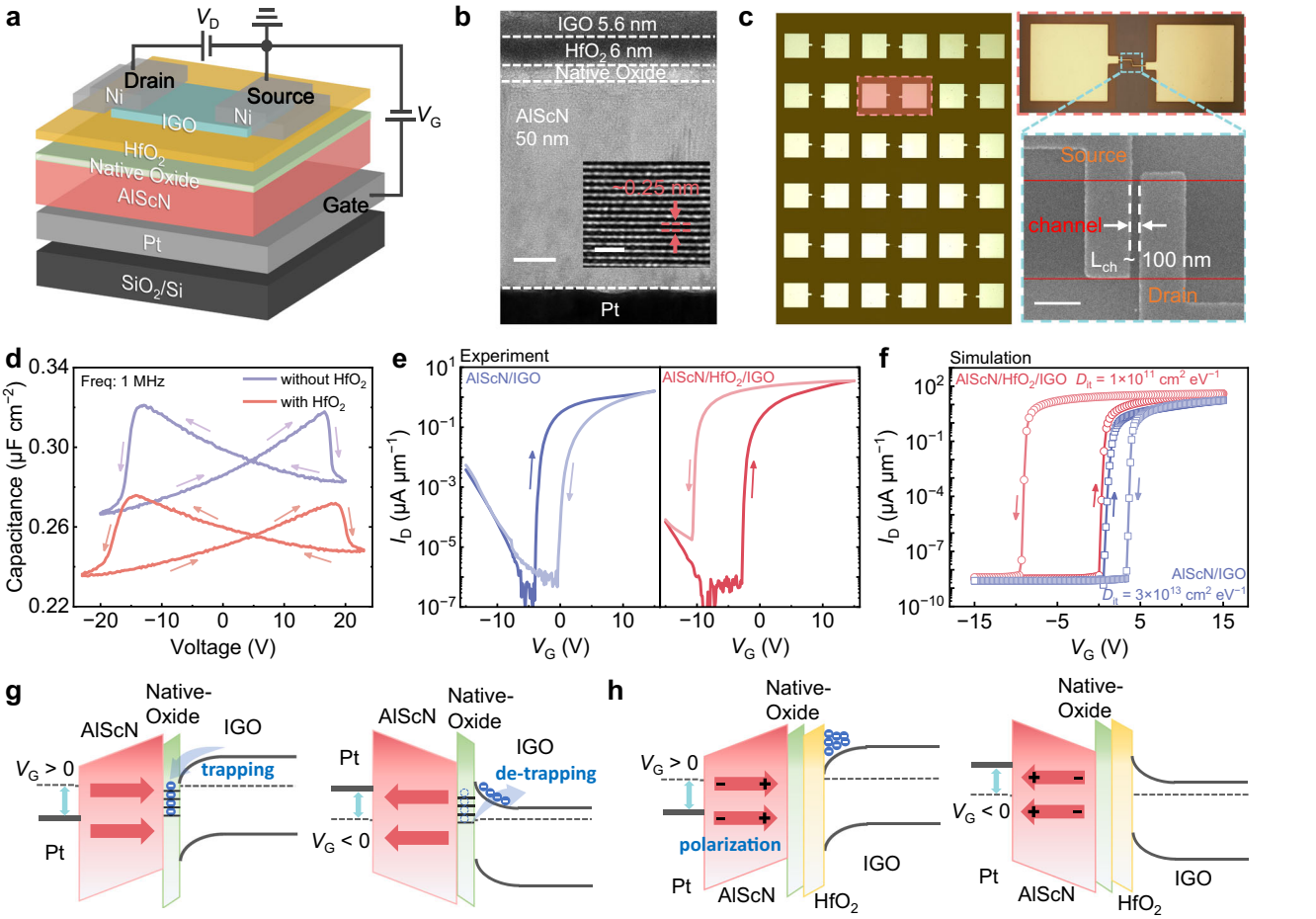


Fig. 2 | Comparison of the $Al_{0.75}Sc_{0.25}N$ MFS and MFIS FeFETs. a Schematic view of a Pt/AlScN/HfO_2/IGO MFIS FeFET. **b** Cross-sectional TEM image of the MFIS gate stack with a scale bar of 10 nm. The inset shows the lattice of the AlScN film with a scale bar of 1 nm. **c** Optical microscopy and SEM images of the FeFET array and a single device. The scale bar in the SEM image is 500 nm. **d** *C*–*V* curves of ex-situ FE capacitors with different gate stacks. The hysteresis direction of the *C*–*V* loops is indicated by arrows. **e**, **f** Experimental and simulation results of transfer characteristic curves for the MFS and MFIS structure devices at a L_{ch} of 2 μm and V_D = 0.05 V. **g**, **h** Energy band diagrams of the MFS and MFIS devices related to the charge dynamics under different gate voltages.

presents an SEM image of a 100-nm MFMIS device. Supplementary Fig. 15 presents the *C*–*V* measurement results of the metal–ferroelectric–metal–insulator–metal (MFMIM) capacitors, which also show a clear butterfly shaped curve representing maintained ferroelectricity. Notably, for the MFIS FeFETs, the transfer characteristic curves exhibit a large memory window (MW) only at the initial state. Upon DC sweeping, starting from the second cycle, V_{TH} undergoes a pronounced negative shift, and the MW decreases sharply to only half of its initial value (8.9 V to 4.6 V). (Fig. 3b). IGO is a typical n-type semiconductor, and its carriers are generally considered to be oxygen vacancies and doped hydrogen[36,56]. Owing to the lack of ambipolar carriers, the electrostatic potential on the IGO channel has a two-dimensional distribution, and the role of the channel of an FeFET cannot be directly regarded as the same as that of the electrode of a ferroelectric capacitor[35,54]. Therefore, during DC-sweep cycling, the V_{TH} of the MFIS-stack devices exhibits a negative shift (Fig. 3b), which causes an undesirable substantial reduction in the MW. This significant variation is unacceptable for stable nonvolatile memory applications. As shown in Fig. 3c, the V_{TH} shift of the MFMIS FeFETs is clearly much smaller than that of the MFIS FeFETs. A stable average MW of 20.6 V is achieved for a 150-nm MFMIS device under 50 DC sweep cycles without obvious variation. The statistical distribution of the V_{TH} during DC cycling is shown in Supplementary Fig. 16a. The ON/OFF ratio of the MFMIS FeFETs is greater than 10^7, indicating excellent gate control capability and low leakage.

At the first DC forward sweep of V_G, the electron concentration in the IGO channel gradually changes from depletion to accumulation. The whole FE layer can be polarized under the electric field from gate to channel (E_{GC}) and maintain the accumulation of the electrons in the channel, corresponding to the programming state. However, for the MFIS FeFETs, when the V_G sweeps backward, the depletion-state channel with a floating potential cannot provide an electric field to switch the FE region under the channel. Polarization switching only occurs under the drain/source contacts (Fig. 3d). Therefore, the V_{TH} of MFIS devices shifts negatively because of the upward polarized FE domains under the channel. In contrast, for the MFMIS FeFETs, the floating metal has a sufficient number of free charges so that the total FE domains can accomplish polarization switching (Fig. 3e). Finally, through a series of systematic optimizations, by integrating sputtered AlScN ferroelectric layers and ALD AOS channels, we ultimately fabricated AlScN FeFETs with excellent performance, which exhibit highly promising potential in memory and neuromorphic computing applications.

To explore the feasibility of high-reliability nonvolatile memory applications, we further measured the memory characteristics of AlScN MFMIS FeFETs. Supplementary Fig. 17 shows the transfer characteristic curves of a 300-nm MFMIS device under different V_G sweep

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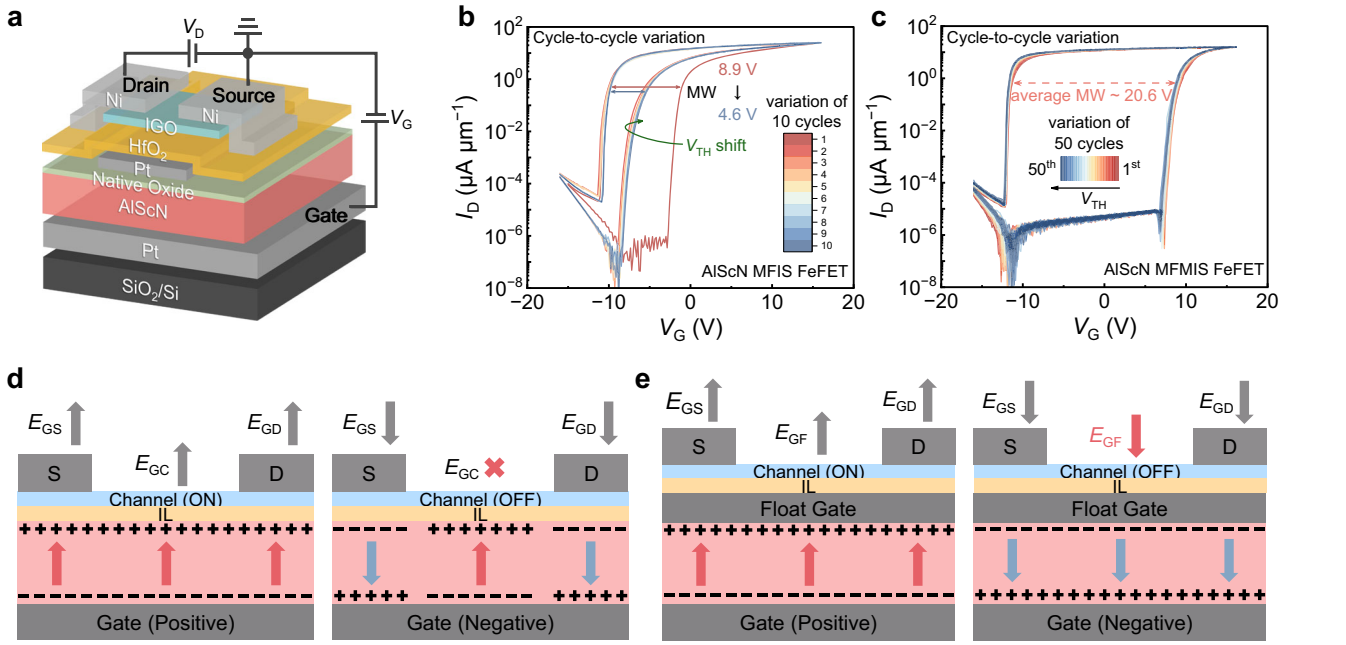


Fig. 3 | Comparison of the $Al_{0.75}Sc_{0.25}N$ MFIS and MFMIS FeFETs. a Schematic view of a Pt/AlScN/Pt/HfO_2/IGO MFMIS FeFET. **b** Cycle-to-cycle variation of the transfer curves of the 200-nm L_{ch} AlScN MFIS FeFET at $V_D = 0.05$ V, showing an obvious negative shift of V_{TH} from the first to the second DC sweep cycles. **c** Cycle-to-cycle variation of the transfer curves of the 150-nm L_{ch} AlScN MFMIS FeFET at $V_D = 0.05$ V, showing a great V_{TH} uniformity under 50 DC sweep cycles. **d**, **e** Schematics of the polarization switching mechanism of MFIS and MFMIS FeFETs.

ranges. The MW decreases as the V_G range is reduced, and all the curves exhibit counterclockwise hysteresis. The MW saturates when $|V_G|$ exceeds 15 V. The device-to-device variation among 30 MFMIS devices with an L_{ch} of 200 nm is shown in Fig. 4a. The statistical distribution of the extracted V_{TH} is presented in Supplementary Fig. 16b. The mean and standard deviation of the MW for the 30 devices are 18.9 V and 0.9 V, respectively, demonstrating the potential for stable large-scale fabrication. Figure 4b presents the transfer characteristics of MFMIS FeFETs with varying L_{ch} obtained under identical V_G. The extracted MW and V_{TH} values are provided in Supplementary Fig. 18. The MW increases from 5.2 V to 20.5 V as L_{ch} decreases from 1 μm to 100 nm. Supplementary Fig. 19 presents the output characteristics of a 100-nm L_{ch} MFMIS FeFET, for which a high current density of 760 μA μm^{-1} at $V_D = 0.8$ V is obtained. The reliability of our devices is also measured via endurance and retention tests. During the endurance test, the source and drain are grounded, and +16/−17 V square-wave program/erase (PGM/ERS) pulse cycles with a pulse width of 10 ms are applied to the gate. As shown in Supplementary Fig. 20a, under this pulse scheme, the device can be switched with an ON/OFF ratio exceeding 10^6, indicating that our endurance test is performed under full-switching conditions. After a specific number of endurance pulses, the V_{TH} of the device in both states is read by a DC dual sweep of I_D–V_G at $V_G = \pm 16$ V and $V_D = 0.05$ V (Supplementary Fig. 20b). The MW evolution of the MFMIS FeFET during 10^6 endurance cycles is shown in Fig. 4c, and the MW began to decrease because of the negative shift in V_{TH+} after 10^5 endurance cycles. The device can still maintain normal operation for up to 10^6 cycles with an MW of 12.8 V. Retention is a key metric for evaluating the data storage capability of devices. A +16/−17 V, 50 ms PGM/ERS pulse is applied at the gate to change the state of the device, followed by reading the drain current after the retention time at $V_D = 0.05$ V with gate grounded. The retention characteristics are shown in Fig. 4d; after the PGM/ERS pulse, our device still maintains an ON/OFF ratio of 10^8 at 10,000 s. When further extrapolated to >10 years, the ON/OFF ratio still exceeds 10^6, indicating great retention of the AlScN MFMIS FeFET. Moreover, we characterized the operating performance of the MFMIS devices at different temperatures, as shown in Fig. 4e. Previous studies have suggested that the E_c of ferroelectric materials generally decreases with increasing temperature[52,57]. Therefore, the MW gradually decreases at low temperatures due to the increased E_c when measured in the same V_G range (Supplementary Fig. 22). At a temperature down to 100 K, our devices still maintain a counterclockwise hysteresis curve and an MW above 1 V. As the temperature increases from room temperature to 400 K, the MW remains approximately stable, accompanied by a gradual increase in the leakage current and a corresponding decrease in the ON/OFF ratio (Supplementary Fig. 23). The devices consistently maintain a counterclockwise hysteresis over the entire investigated temperature range, further proving that the hysteresis of MFMIS devices is dominated by ferroelectricity instead of thermally activated charge trapping. The stable and reliable AlScN MFMIS FeFETs with large MWs also possess excellent capability for multibit operation. Supplementary Fig. 21 presents a 4-bit retention of 100 s after various PGM pulses with 16 distinct states. Figure 4f shows the 64 distinct curves corresponding to different V_{TH} states, which are obtained by applying various PGM pulses with amplitudes ranging from 13 to 17 V and a pulse width of 50 ms, indicating the feasibility of 6-bit operation of our MFMIS FeFETs. Figure 4g, h show the benchmarks of multibit operation capability versus normalized MW and endurance versus retention from reported AlScN-based FeFETs, respectively[30,31,37,38,58–62]. More information on the benchmark is provided in Supplementary Table 3. As evident from the figures, compared with those of other AlScN-based FeFET works, the normalized MW of 0.41 V nm^{-1}, 6-bit operation, a retention of 10 years and an endurance of 10^6 cycles highlight the potential for implementing high-performance AlScN MFMIS FeFETs in large-scale reliable memory applications.

Neural network application of the devices

The AlScN FeFETs with large MWs and multibit operation capabilities can also simulate the long-term potentiation (LTP) and long-term depression (LTD) behaviors of biological synapses, which is attributed to the accurate modulation of channel conductance by ferroelectric

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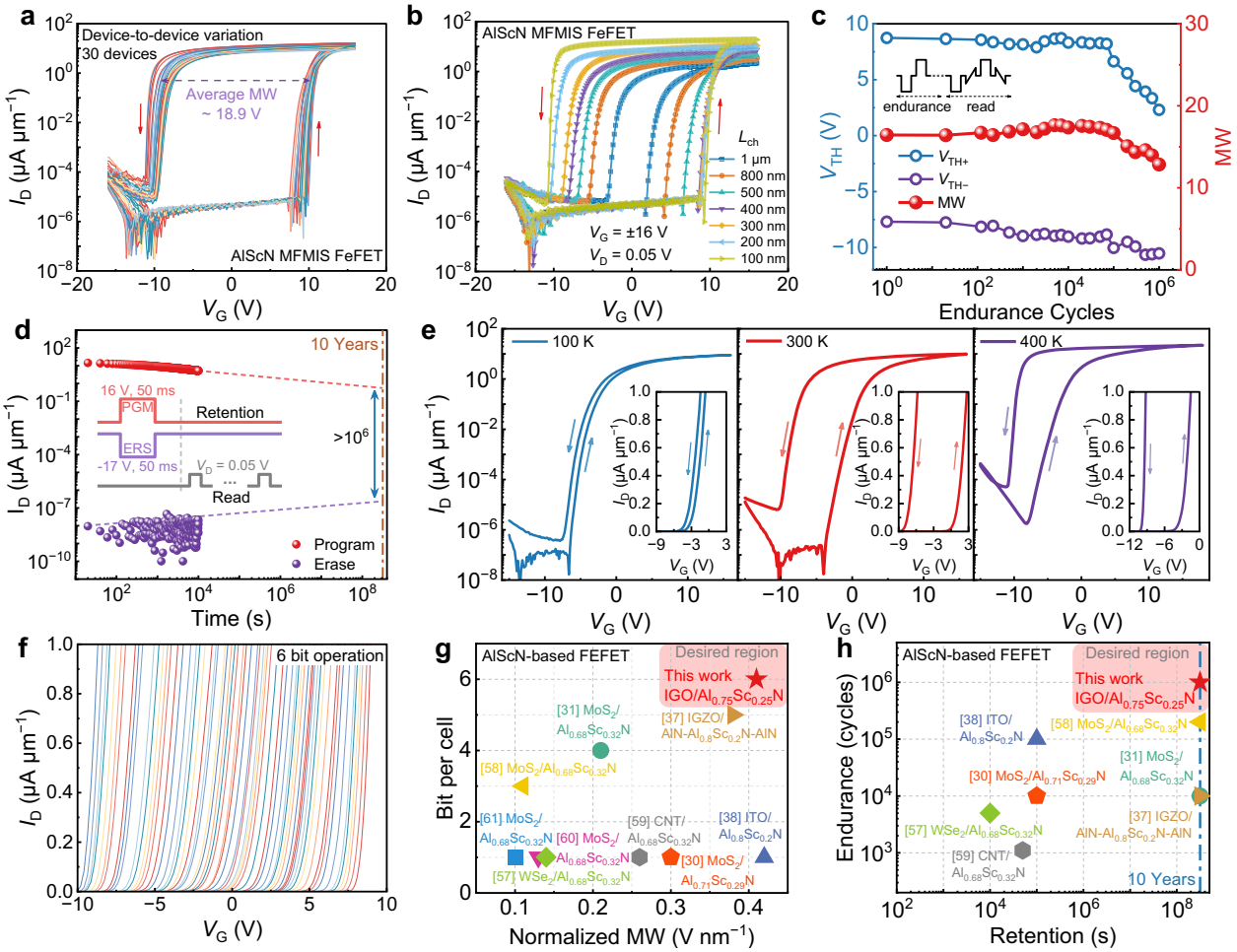


Fig. 4 | Memory characteristics of the $Al_{0.75}Sc_{0.25}N$ MFMIS FeFETs. a Device-to-device variation of the transfer curves of the 200-nm L_{ch} AlScN MFMIS FeFETs at $V_G = \pm 16$ V and $V_D = 0.05$ V. **b** Transfer characteristics of AlScN MFMIS FeFETs with various L_{ch} at $V_G = \pm 16$ V and $V_D = 0.05$ V. **c** Endurance characteristics of a MFMIS FeFET under PGM/ERS pulses of 16 V/−17 V, 10 ms with a L_{ch} of 300 nm. **d** Retention characteristics of MFMIS FeFETs under a PGM/ERS pulse of +16 V/−17 V, 50 ms with a read pulse of $V_D = 0.05$ V. **e** Transfer characteristics in both logarithmic and linear scales of 800-nm L_{ch} MFMIS FeFETs at different temperatures. **f** Multibit operation of MFMIS FeFETs, showing 64 distinct states for 6-bit operation. **g** Benchmark of multibit operation versus normalized MW from reported AlScN-based FeFETs. **h** Benchmark of endurance versus retention from reported AlScN-based FeFETs.

polarization via a series of positive and negative pulses. Figure 5a presents the conductance responses of the AlScN FeFETs during 21 repetitive LTP/LTD test cycles with pulses of ±16 V, 250 μs. Each cycle consists of 128 continuous positive pulses and negative pulses. Figure 5b presents the cycle-to-cycle variation on average of ~4.2%, which shows slight fluctuations and favorable repeatability. To demonstrate the potential application of our AlScN FeFET synapse in neuromorphic computing, we used it to recognize handwritten digits in the Modified National Institute of Standards and Technology (MNIST) database. The nearly linear LTP/LTD characteristics are utilized to modulate the synaptic weight updates. We constructed a fully connected neural network model with a 3-layer network for handwritten digit recognition (Fig. 5c). The network consists of three fully connected layers: an input layer with 400 neurons corresponding to the 20 × 20 pixels in one image, a hidden layer with 155 neurons and an output layer with 10 neurons for the recognition of digits 0–9. Figure 5d shows a color map of the output layer's weight distribution before and after training (quantized to ±127 levels by using differential pairs of 128-state devices). The size of the colormap is 20 × 77. The weights are transformed from a random distribution to a structured distribution. These results suggest that AlScN FeFET synapses are sufficient for a neuron to store a 7-bit quantized weight. The experimentally measured LTP/LTD characteristics are used as discrete conductance-update states in the neuromorphic simulation. To implement signed synaptic weights, we used a differential-pair representation, where the effective weight is determined by the difference in conductance between two FeFET devices. Therefore, the numerical weight range used in the network is obtained from normalized differential conductance rather than from positive physical conductance in a single device. Details of the conductance-to-weight mapping and training procedure are provided in Methods and Supplementary Note 2. After the neural network completes 20 training epochs against 60,000 handwritten images, we perform a recognition test using 10,000 handwritten images. Figure 5e shows the recognition confusion matrix for the 10,000 images test. The digits of 5 and 8 have lower accuracies of 89.5% and 87.8%, respectively, because of shared features. This may be addressed by adaptively adjusting the training parameters or optimizing the neural network architectures. The relationships between the recognition accuracy rate and training epochs of the software and different quantized FeFETs are shown in Fig. 5f. The recognition accuracy rate increases rapidly within the first 5 training epochs because enough large training images exist in the batch. The average recognition accuracy of our neural network simulation model reaches 93.02% with 7 bits (128 states) of FeFET, approaching the ideal 97.93% with software implementation. As the trained weights are re-quantized from 7 bits to 5 bits, 3 bits and 2 bits in our FeFET synapse, the recognition accuracy

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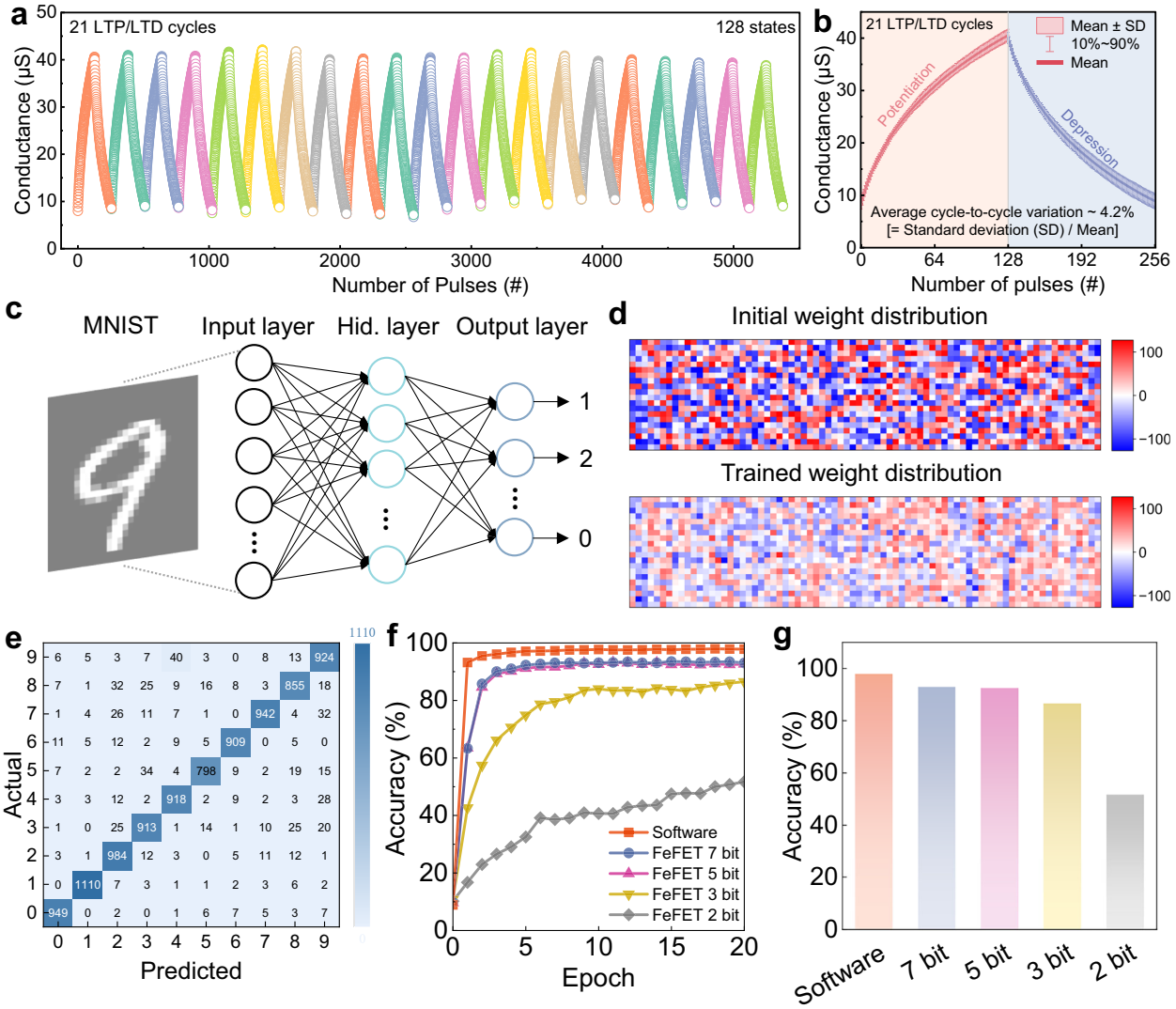


Fig. 5 | Neural network application of $Al_{0.75}Sc_{0.25}N$ MFMIS FeFETs. **a** Conductance responses of the AlScN MFMIS FeFETs during repetitive LTP/LTD measurements of 21 cycles. Each cycle of potentiation and depression has 128 different conductance states. **b** Cycle-to-cycle variation of 21 LTP/LTD cycles from (**a**). The box limits represent the mean ± standard deviation (SD). The whiskers indicate the 10th–90th percentile range. The solid line connects the mean values. The average variation (SD/Mean) across the 21 LTP/LTD cycles is 4.2%. **c** Neuromorphic network with three layers, each containing 400 input neurons, 155 hidden neurons, and 10 output neurons. **d** Color map before and after training the weights of the output layer. The size of the colormap is 20 × 77. **e** Confusion matrix of the test results under 20 training epochs, with 60,000 images used for training and 10,000 images used for testing. The outputs 0–9 denote the desired output handwritten digits. **f** Accuracy of handwritten digit recognition for different numbers of epochs. The accuracy of the 7-bit FeFET is 93.02%, alongside the accuracy of the software, and the FeFET re-quantized 5 bits, 3 bits, and 2 bits. **g** Benchmark accuracy of handwritten digit recognition with software, the FeFET re-quantized 7 bits, 5 bits, 3 bits, and 2 bits.

decreases from 93.02% to 92.63%, 86.55% and 51.71%, respectively (Fig. 5g). To further evaluate the neuromorphic performance, we compared the MNIST recognition accuracy of our AlScN MFMIS FeFET synapse with that of representative ferroelectric synaptic devices, including high-impact reports and AlScN-related synaptic devices (Supplementary Table 4). The proposed device achieves an accuracy of 93.02%, which is comparable to that of these representative ferroelectric synaptic devices. In addition to recognition accuracy, our device provides an annealing-free AlScN process, a large memory window, a stable multilevel operation, and strong memory reliability, indicating its potential for BEOL-compatible neuromorphic computing.

In conclusion, we report high-performance BEOL-compatible annealing-free FeFETs based on wurtzite AlScN ferroelectric films and ultrathin IGO channels. A high-quality HfO_2 IL effectively decreases the number of interface trap states between AlScN and IGO. The optimized MFMIS-FeFETs deliver a large MW of 20.6 V, a high ON/OFF ratio of more than 10^7, and a high current density of 760 μA μm^{-1}. Reliability tests demonstrate excellent endurance (10^6 program/erase cycles with MW > 12.8 V) and retention (extrapolated to >10 years with ON/OFF ratios > 10^6), along with stable ferroelectricity over a wide range of temperatures. Moreover, the device supports a 6-bit operation with 64 distinct V_{TH} states. These competitive characteristics underscore the potential of wurtzite AlScN-based FeFETs as versatile platforms for next-generation NVM and neuromorphic computing systems.

Methods

In situ AlScN capacitor fabrication

A 36-nm Pt bottom electrode (BE) was deposited on SiO_2/Si substrates via DC sputtering with a 2-inch Pt metal target. The sputtering process was conducted at room temperature with a power of 60 W, a working pressure of 2 mTorr and an Ar gas flow of 20 sccm. A 50-nm

$Al_{0.75}Sc_{0.25}N$ film was subsequently deposited via reactive sputtering using 2-inch Al and Sc metal targets at 300 °C under a working pressure of 2 mTorr with an Ar gas flow of 10 sccm. Additionally, the N_2 samples used pure N_2, and the N_2/H_2 samples used 5% H_2/95% N_2 as the reactive gas with the same flow of 6 sccm. The normal DC power for the Al target was 130 W, whereas the pulse DC power for the Sc target was 100 W at 150 kHz and a duty ratio of 30%. Following the deposition of AlScN, a Pt top electrode (TE) was sputtered in situ under the same conditions as the BE without breaking the chamber vacuum. All the sputtering processes were carried out in a sputtering system with a base pressure below 10^{-8} Torr. Finally, the TE was patterned via electron-beam lithography (EBL) and etched via ion beam etching (IBE) to form a circular electrode with a radius of 30 µm.

Transistor and ex situ capacitor fabrication

The Pt BE and AlScN films were sputtered via the same processes as those used for the N_2/H_2 samples. A 5.6-nm-thick IGO channel was deposited via ALD at 225 °C. $Ga(CH_3)_3$ (TMGa) and O_3 were employed as the precursors for Ga_2O_3, whereas $In(CH_3)_3$ (TMIn) and H_2O were used as the precursors for In_2O_3. The ALD cycle ratio of In_2O_3 to Ga_2O_3 in the IGO channel was maintained at 1:1. The channel region was defined by EBL, followed by isolation by etching with hydrochloric acid. The channel width was fixed at 1 µm. For the MFIS FeFETs and MFIM capacitors, a 6-nm HfO_2 interlayer was deposited via ALD at 90 °C using $Hf[N(C_2H_5)CH_3]_4$ (TEMAHf) and H_2O as precursors or 250 °C using TEMAHf and O_3 as precursors. For the MFMIS FeFETs and MFMIM capacitors, 20-nm Pt floating metal was deposited via DC sputtering at room temperature with a power of 20 W, a working pressure of 2 mTorr and an Ar gas flow of 20 sccm. All the source/drain contacts of FETs and TEs of ex-situ capacitors with different gate stacks were patterned via EBL and deposited via e-beam evaporation (EBE) via a lift-off process. There was no annealing process in any of the FeFETs or capacitors.

Device characterization

The surface morphology images of the AlScN and IGO films were obtained via AFM (Oxford Cypher S). The crystallization of Pt and AlScN was characterized via X-ray diffraction (XRD) (Bruker D8 Discover). The thickness and elemental composition of each layer were determined via high-resolution TEM and EDS line scanning, respectively (Tecnai G2 F20). The atomic ratio of the IGO channel was determined via XPS (AXIS SUPRA+). The channel length was measured via SEM (Raith eLINE Plus). Electrical measurements (*C–V* curves and transfer and output characteristics) were performed via Keysight's B1500A semiconductor parameter analyzer, and SPGU modules were used for the FeFET pulse tests. Ferroelectric measurements (*P–E*, *J–E* and PUND tests) were carried out via a Radiant Precision Premier II ferroelectric tester. The devices were measured under vacuum in an electrically shielded probe stage (Lakeshore TTPX).

Test configurations of different devices are presented in Supplementary Fig. 2. For the ferroelectric dynamic tests such as *P–E*, *J–E*, and PUND measurements of the MFM capacitors, the driving voltage was applied to the top electrode while the bottom electrode was grounded as shown in Supplementary Fig. 2a. The top electrode was circular with a radius of 30 µm. Triangular waveforms with different frequencies and amplitudes were used as the test waveforms. For the DC or quasi-static tests such as *C–V* loop measurements of capacitors with different stacks, the measurement configuration was consistent with that of the FeFETs: the driving voltage was applied to the bottom electrode (similar to gate) and the top electrode (similar to source/drain) was grounded as shown in Supplementary Fig. 2b, with the top electrode being circular and having a radius of 25 µm. The *C–V* measurements were performed at a frequency of 1 MHz, with the test signal consisting of a DC bias superimposed with an AC small signal of 30 mV.

Neuromorphic simulation

In the neuromorphic simulation, each signed synaptic weight was represented by a differential pair of FeFET devices. The effective weight was defined as $W_{eff} = \alpha(G^+ - G^-)$, where G^+ and G^- are the conductance states of the two FeFET devices and α is a normalization factor. This representation allows both positive and negative numerical weights to be encoded using physically positive device conductance states. The experimentally measured LTP/LTD conductance values were used as discrete update states during training; thus, the simulation incorporated the finite conductance range and non-ideal update behavior of the measured devices.

Data availability

Relevant data supporting the key findings of this study are available within the article and the Supplementary Information file. All raw data generated during the current study are available from the corresponding authors upon request. Source data are provided with this paper.

Code availability

The code for this study is available from the corresponding author.

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Author contributions

X. L. proposed and supervised the project. K. W. designed the experiments, and carried out the device fabrication, measurements, and data analysis. T. G., H. Liu., S. C., and Y.C. carried out the neuromorphic computing simulation. K. H. conducted and discussed the theoretical mechanism analysis. H. Liu performed the simulation of the FeFETs. K. H., H. Liang, and J. Z. advised and assisted in the growth of the AlScN film. C.Y. and J. H. advised and assisted in the growth of the IGO film. Z.Q. and all the other authors discussed the results and wrote the paper.

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Competing interests

The authors declare no competing interests.

Additional information

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